https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	x is the independent variable and *h* is the dependent variable **or** vary *x* and measure/determine *h*	**1**
	keep *r* and/or *k* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • labelled spring • upper ball vertically (by eye) above the spring and at least one of the balls labelled • vertical (by eye) rule at least from top of spring to position of upper ball	**1**
	method to determine *x*, e.g. use a rule/calipers to measure original and final length of spring and find the difference	**1**
	use of micrometer/calipers/rule to measure diameter of ball	**1**
	method to determine *h* described, e.g. measure the distance between the top of the ball at maximum height and the top of the ball on the spring **or** measure the distance between the bottom of the ball at maximum height and the bottom of the ball on the spring	**1**
	Method of analysis	
	plot a graph of *h* against x^2 (or lg *h* against lg *x*)	**1**
	relationship valid if a straight line through (0,0) (for lg *h* against lg *x* straight line with gradient = 2)	**1**
	$\rho = \dfrac{3k}{8\pi r^3 g \times \text{gradient}}$ $\left(\text{for lg}\,h \text{ against lg } x, \quad \rho = \dfrac{3k}{8\pi g r^3 \times 10^{y\text{–intercept}}}\right)$	**1**

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Question	Answer		Marks
1	**Additional detail including safety considerations**		
	D1	use large box/cage to collect ball (to prevent ball rolling on floor/bouncing) **or** reasoned method to avoid draughts, e.g. switch off fans, close windows, use a screen	**6**
	D2	expression to determine k from relevant experiment, e.g. $k = mg / x$ or gradient of F–x graph	
	D3	stand on bench with clamped rule vertically to measure vertical distance	
	D4	method to ensure clamped rule to measure h is vertical, e.g. correctly positioned set square indicated at right angles between the rule and the horizontal surface **or** plumb line shown in appropriate position	
	D5	$r = d / 2$ when diameter measured	
	D6	repeat diameter measurement in different directions and find average	
	D7	repeat experiment for each value of x and determine average h	
	D8	method to securely fix spring to the bench e.g. tape/G-clamp	
	D9	experiment to determine k, e.g. place mass m on the spring and measure compression x	
	D10	video (camera) shown level (by eye) with elevated ball and description of play back frame by frame or slow motion	

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